

**BISCHOFSHOFEN, Austria**

WMO: 111410

Lat: 47.407N

Lon: 13.221E

Elev: 548

StdP: 94.92

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

**Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |             | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-12.4 | (c)<br>-10.4 | (d)<br>-14.1                  | (e)<br>1.2 | (f)<br>-11.8 | (g)<br>-12.0 | (h)<br>1.4 | (i)<br>-9.8 | (j)<br>5.1            | (k)<br>-1.0 | (l)<br>4.4 | (m)<br>-1.1 | (n)<br>1.1            | (o)<br>160 | (p)<br>0.509 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |           |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|-----------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |           |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD      |
| (a)<br>7      | (b)<br>10.4            | (c)<br>29.8     | (d)<br>19.3 | (e)<br>27.8 | (f)<br>18.6 | (g)<br>26.0 | (h)<br>17.8 | (i)<br>20.2         | (j)<br>27.6 | (k)<br>19.3 | (l)<br>26.0 | (m)<br>18.5 | (n)<br>24.4 | (o)<br>2.9           | (p)<br>10 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.9                            | 13.7 | 22.0 | 17.1 | 13.0 | 21.4 | 16.4 | 12.5 | 20.5 | 60.3          | 27.7 | 57.3 | 26.0 | 54.6 | 24.5 | 25.8              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 5.9               | 4.9  | 4.1 | -14.6                      | 33.6 | 3.2                | 1.5 | -16.9                                              | 34.7 | -18.8      | 35.6 | -20.6      | 36.5 | -22.9      | 37.6 |  |  |
|                   |      |     | -15.0                      | 22.2 | 3.0                | 1.1 | -17.1                                              | 23.0 | -18.9      | 23.6 | -20.6      | 24.2 | -22.7      | 25.0 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.9    | -2.3  | 0.2   | 4.8   | 9.7   | 13.7  | 16.9  | 18.5  | 18.2  | 14.1  | 9.8   | 3.7   | -0.9  |
|                                                                        | DBStd       | 8.11   | 3.75  | 3.72  | 3.80  | 3.88  | 3.54  | 3.78  | 3.31  | 3.17  | 3.21  | 3.45  | 3.70  | 3.47  |
|                                                                        | HDD10.0     | 1462   | 382   | 275   | 163   | 52    | 9     | 2     | 0     | 1     | 5     | 46    | 190   | 338   |
|                                                                        | HDD18.3     | 3552   | 640   | 508   | 418   | 259   | 147   | 70    | 39    | 41    | 129   | 264   | 438   | 597   |
|                                                                        | CDD10.0     | 1069   | 0     | 0     | 2     | 43    | 124   | 209   | 264   | 255   | 129   | 41    | 2     | 0     |
|                                                                        | CDD18.3     | 117    | 0     | 0     | 0     | 0     | 4     | 28    | 45    | 37    | 3     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 1100   | 0     | 0     | 0     | 11    | 82    | 289   | 380   | 305   | 33    | 1     | 0     | 0     |
|                                                                        | CDH26.7     | 292    | 0     | 0     | 0     | 1     | 12    | 80    | 113   | 83    | 3     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 1.4    | 1.2   | 1.4   | 1.6   | 1.7   | 1.7   | 1.7   | 1.6   | 1.4   | 1.3   | 1.3   | 1.2   | 1.2   |
| (15) Precipitation                                                     | PrecAvg     | 1342   | 71    | 68    | 94    | 85    | 128   | 164   | 185   | 171   | 124   | 92    | 84    | 76    |
|                                                                        | PrecMax     | 1552   | 162   | 182   | 198   | 145   | 184   | 238   | 281   | 246   | 198   | 150   | 164   | 215   |
|                                                                        | PrecMin     | 1099   | 9     | 20    | 38    | 23    | 28    | 111   | 91    | 105   | 58    | 21    | 12    | 23    |
|                                                                        | PrecStd     | 105    | 39    | 40    | 44    | 33    | 42    | 35    | 52    | 36    | 37    | 37    | 33    | 38    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 6.7   | 11.7  | 19.0  | 24.8  | 28.4  | 31.6  | 32.3  | 31.5  | 26.5  | 21.8  | 17.8  | 8.2   |
|                                                                        |             | MCWB   | 4.8   | 6.1   | 9.9   | 13.6  | 17.1  | 19.6  | 20.2  | 19.7  | 17.8  | 15.0  | 10.7  | 6.1   |
|                                                                        | 2%          | DB     | 4.2   | 8.5   | 16.1  | 22.2  | 25.7  | 29.0  | 29.7  | 29.1  | 24.0  | 19.2  | 12.4  | 5.5   |
|                                                                        |             | MCWB   | 2.9   | 4.5   | 8.9   | 12.5  | 15.9  | 18.8  | 19.7  | 19.4  | 16.7  | 13.7  | 9.0   | 4.4   |
|                                                                        | 5%          | DB     | 2.9   | 6.2   | 13.5  | 19.8  | 23.5  | 26.8  | 27.5  | 26.8  | 21.9  | 17.1  | 10.1  | 4.1   |
|                                                                        |             | MCWB   | 2.0   | 3.3   | 7.7   | 11.5  | 14.8  | 17.8  | 18.9  | 18.7  | 15.9  | 12.5  | 7.8   | 3.2   |
|                                                                        | 10%         | DB     | 1.9   | 4.5   | 10.8  | 17.5  | 21.1  | 24.4  | 25.2  | 24.6  | 19.8  | 15.0  | 8.3   | 3.1   |
|                                                                        |             | MCWB   | 1.3   | 2.6   | 6.6   | 10.4  | 13.7  | 17.0  | 18.0  | 17.9  | 14.9  | 11.7  | 6.8   | 2.3   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 5.0   | 6.8   | 10.8  | 14.5  | 18.2  | 20.9  | 21.6  | 21.1  | 18.7  | 15.6  | 11.2  | 6.4   |
|                                                                        |             | MCDB   | 6.2   | 10.8  | 17.2  | 23.3  | 26.1  | 29.0  | 29.1  | 28.8  | 24.3  | 20.2  | 16.3  | 7.9   |
|                                                                        | 2%          | WB     | 3.1   | 4.9   | 9.3   | 13.0  | 16.7  | 19.5  | 20.3  | 20.0  | 17.5  | 14.2  | 9.3   | 4.7   |
|                                                                        |             | MCDB   | 4.0   | 7.7   | 15.0  | 20.9  | 23.9  | 27.0  | 27.9  | 27.0  | 22.6  | 17.9  | 12.1  | 5.5   |
|                                                                        | 5%          | WB     | 2.1   | 3.7   | 8.1   | 11.9  | 15.5  | 18.5  | 19.4  | 19.1  | 16.5  | 13.1  | 8.1   | 3.3   |
|                                                                        |             | MCDB   | 2.7   | 5.7   | 12.6  | 18.4  | 21.8  | 25.1  | 26.1  | 25.2  | 20.8  | 16.1  | 9.9   | 4.0   |
|                                                                        | 10%         | WB     | 1.4   | 2.7   | 6.8   | 10.8  | 14.5  | 17.5  | 18.5  | 18.2  | 15.4  | 12.0  | 6.9   | 2.4   |
|                                                                        |             | MCDB   | 1.9   | 4.2   | 10.4  | 16.5  | 19.8  | 23.1  | 24.0  | 23.7  | 18.8  | 14.5  | 8.3   | 3.0   |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 5.5   | 7.6   | 9.9   | 11.3  | 11.0  | 10.5  | 10.4  | 10.4  | 9.5   | 9.4   | 6.2   | 5.2   |
|                                                                        | 5% DB       | MCDBR  | 5.2   | 10.6  | 15.1  | 16.6  | 16.0  | 15.7  | 15.1  | 15.1  | 13.3  | 12.4  | 9.5   | 6.0   |
|                                                                        |             | MCWBR  | 4.0   | 6.8   | 8.1   | 7.6   | 6.8   | 6.3   | 5.9   | 6.1   | 6.5   | 7.0   | 6.0   | 4.5   |
|                                                                        | 5% WB       | MCDBR  | 4.7   | 9.3   | 14.0  | 15.2  | 14.3  | 14.1  | 13.7  | 13.6  | 12.0  | 11.4  | 9.0   | 5.4   |
|                                                                        |             | MCWBR  | 3.7   | 6.0   | 7.7   | 7.1   | 6.4   | 5.9   | 5.6   | 6.0   | 6.1   | 6.6   | 5.8   | 4.1   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.290 | 0.322 | 0.371 | 0.415 | 0.430 | 0.437 | 0.437 | 0.428 | 0.390 | 0.357 | 0.323 | 0.291 |
|                                                                        | taud        |        | 2.357 | 2.262 | 2.198 | 2.167 | 2.188 | 2.221 | 2.227 | 2.270 | 2.347 | 2.406 | 2.432 | 2.417 |
|                                                                        | Ebn at Noon |        | 792   | 832   | 839   | 835   | 836   | 830   | 824   | 815   | 814   | 787   | 747   | 749   |
|                                                                        | Edh at Noon |        | 71    | 97    | 121   | 138   | 141   | 137   | 134   | 124   | 104   | 84    | 67    | 60    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.30  | 2.24  | 3.21  | 4.10  | 4.68  | 5.18  | 5.27  | 4.69  | 3.43  | 2.40  | 1.35  | 1.06  |
|                                                                        | RadStd      |        | 0.16  | 0.28  | 0.38  | 0.52  | 0.56  | 0.54  | 0.51  | 0.56  | 0.46  | 0.25  | 0.22  | 0.17  |

**Historical Trends**

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (91 neighbors) |  | +0.44 | N/A     | N/A    | N/A     | +0.40 | +0.34 | N/A         | N/A     | +70     | +33     |

Nomenclature: See separate page